

CI/CD, GitOps, Secret Management

and other Buzzwords

Motivation

- Monitoring many different Git Repos, branches etc. is annoying and error prone
- Secrets must be (re)used in many parts of the system
- Building can be painful when working on different chip architectures

Secret Management

- Using `HashiCorp Vault` to store all our secrets across different environments, applications etc.
- Integrates seamlessly with other tools we plan to use (following slides)
- Deployed on separate VM to be easily accessible by all systems (Dev Laptops, k8s cluster)
- Integrates great with CI/CD pipelines -> no longer wasting time searching for Auth-Tokens

Continuous Integration

- Using GitHub Actions
- Free Tier available that should meet our needs
 - 2000 mins/month free building in the cloud on Linux builders
- Build artifacts are consistent across developers
- There is a unified current version always on `main`

GitOps / Continuous Deployment

- In connection with previous slide
- Using `Fleet` to monitor a unified git repo with all helm resources that are deployed on our k8s cluster
- When changes are made to main (e.g. image tag change)
 - Fleet reads the changes
 - Reconciles the state of the cluster to meet the new demands
 - Integrates with secret management
- Advantage: Clear current state of the cluster